Nikita Kazeev

kazeevn@gmail.com · Website · GitHub · LinkedIn · Google Scholar · ORCID

Research Scientist at the intersection of AI and Physics

Work Experience

Perfomax *AI Advisor* **2026–present**

National University of Singapore *Postdoc under Kostya Novoselov* **2022–present**

Leadership

- Program Chair for the AI4X 2025/6 conferences
- Main organiser of the ICLR 2025 Workshop on Machine Learning Multiscale Processes

Wyckoff Transformer, a generative model for materials ICML 2025 · GitHub

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference by 4 orders of magnitude, to 0.05 GPU ms per structure
- Implemented the model in PyTorch; production-ready package published on PyPI
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

⟨CERN | Yandex → HSE University⟩ *Researcher* **2014–2022**

2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.

Generative models uncertainty estimation paper · conference · code 1 · code 2

- Developed the first methods for estimating uncertainty of conditional GANs.
- Led a team of 3 students.

Generative models for fast simulation paper 1 · paper 2 · paper 3 · code

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately 10^5 speed-up compared to the ab-initio simulator.

Education

HSE University & Sapienza Università di Roma **2016–2020**
PhD in Computer Science and Physics (Double Degree)
Supervisors: Andrey Ustyuzhanin & Barbara Sciascia
Thesis: Machine Learning for particle identification in the LHCb detector

Product Management course at Yandex **2018**
Focused coursework in product strategy and execution.

Yandex School of Data Analysis **2013–2015**
Master's-level CS programme
Algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology
MS in Physics **2014–2016**
Optimisation of data processing of the LHCb experiment.
BS in Physics **2010–2014**
Wave-packet molecular dynamics of electrons in nonideal plasma.

International Junior Science Olympiad (IJSO) **Korea, 2008**
Silver medal

Mentorship

- Mentored 9 students (1, 2, 3, 4, 5, 6, 7, 8, 9), 3 interns, and student workshop projects (1, 2).
- Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
- Co-PI of a $3.4m AI Singapore grant on multiscale machine learning.

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage (1, 2), blog posts (1, 2), a science slam, interviews (1, 2), and a public talk.
- Lecturer in the award-winning Coursera Advanced Machine Learning specialisation
- Delivered more than 20 conference talks.

Service

- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

- **Generative modelling for scientific data** — GANs, autoregressive transformers, and diffusion; from fast detector simulation to symmetry-aware crystal generation.
- **Uncertainty quantification** — built the first methods for estimating the uncertainty of conditional GANs; calibrated generative and discriminative models.
- **ML on structured, non-text data** — tabular, graph, and atomistic representations, spanning gradient boosting to transformers.
- **Research leadership** — lead multidisciplinary teams, chair conferences, and mentor students from idea to publication.

ML & AI • Python • PyTorch • HPC • Physics • Research Writing & Speaking • Leadership & Mentorship

Presenting at ICML 2026

July 11, 12:15–13:30 — **Nikita Kazeev** and Ian Babich. "Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments". In: *Forecasting as a New Frontier of Intelligence Workshop at ICML 2026*.

July 11, 15:00–16:30 — **Nikita Kazeev** and Phan Bui Nhat Huyen. "Position: Align AI to Our Aspirations, Not Our Flaws". In: *Pluralistic Alignment Workshop at ICML 2026*.

July 11, 15:30–17:00 — **Nikita Kazeev** and Andrey Ustyuzhanin. "The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories". In: *Statistical Frameworks for Uncertainty in Agentic Systems Workshop at ICML 2026*.

July 11, 11:40–13:30 & 16:10–17:00 — **Nikita Kazeev** and Andrey Ustyuzhanin. "Uncertainty Quantification for LLM Agents via Semantic Abstraction Trajectories". In: *Structured Probabilistic Inference & Generative Modeling Workshop at ICML 2026*.

Selected Publications

Nikita Kazeev, Wei Nong, Ignat Romanov, Ruiming Zhu, Andrey E Ustyuzhanin, Shuya Yamazaki, and Kedar Hippalgaonkar. "Wyckoff Transformer: Generation of Symmetric Crystals". In: *Proceedings of the 42nd International Conference on Machine Learning*. Ed. by Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu. Vol. 267. Proceedings of Machine Learning Research. PMLR, 13–19 Jul 2025, pp. 29495–29526.

Denis Derkach, **Nikita Kazeev**, Fedor Ratnikov, Andrey Ustyuzhanin, and Alexandra Volokhova. "Cherenkov detectors fast simulation using neural networks". In: *Nuclear Instruments and Methods in Physics Research Section A* (2019). **Alphabetic order, I'm the corresponding author.**